

Trade Study TS-001

Portable Power System Selection for Ground Control Station

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

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1. Objective

The objective of this trade study is to evaluate candidate portable power architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the solution that best satisfies the system's operational, technical, and cost requirements. The GCS is being developed as a supporting subsystem of the TOMTOM ISR demonstrator and is responsible for mission planning, vehicle command and control, telemetry monitoring, payload management, and operator situational awareness.

The GCS hardware is being constructed by repurposing the internal components of an ASUS G73S laptop and integrating them into a vintage portable instrument case to create a rugged, portable field computing platform. This custom enclosure is intended to house the laptop motherboard, display, power distribution system, communications equipment, and future mission support hardware while providing improved portability and field usability compared to the original laptop configuration.

Because the Ground Control Station is a subsystem of the larger TOMTOM system, its design must remain consistent with the operational objectives, system requirements, and architectural constraints established for TOMTOM. The selected power solution must therefore provide sufficient endurance for field operations, support future expansion of mission equipment, enable reliable operation under portable deployment conditions, and maintain compatibility with planned system enhancements such as integrated telemetry radios, external peripherals, and renewable energy charging capabilities. In addition to meeting technical performance objectives, the selected architecture should provide an appropriate balance between cost, reliability, maintainability, portability, and future scalability while supporting the overall development goals of the TOMTOM project.

Assumptions

- The ASUS G73S motherboard remains the primary computing platform.
- The GCS will operate independently of commercial power for a minimum of two hours.
- The power system will support future integration of telemetry radios and additional mission equipment.
- The design will be installed within the selected portable instrument case.

Constraints

- Prototype budget shall not exceed \$200 for the initial power subsystem.
- The architecture shall preserve compatibility with future solar charging.
- Components shall be commercially available.

2. Design Requirements

Requirement	Value
Runtime	≥ 2 hours
Operating Voltage	19 VDC laptop input
Budget	<\$200 preferred
Rechargeable	Yes
Field Portable	Yes
Future Solar Integration	Desired
Safe Operation	Required

3. Alternatives Considered

Alternative A

OEM ASUS Laptop Battery

Advantages

- Integrated charging
- No additional electronics
- Lightweight

Disadvantages

- Currently does not work and would necessitate purchasing a replacement to be initially considered

- Limited runtime
- Difficult to replace in custom enclosure
- No solar charging capability

Alternative B

12V Sealed Lead Acid (SLA)

Advantages

- Lowest cost
- Widely available
- Simple charging
- High current capability
- Rugged and durable

Disadvantages

- Heavy
- Lower cycle life
- Lower energy density

Alternative C

12.8V LiFePO₄ Battery

Advantages

- Comparably lightweight
- Long cycle life
- Excellent energy density
- Solar compatible

Disadvantages

- Highest purchase cost

Alternative D

Commercial Portable Power Station

Advantages

- Plug-and-play
- Built-in inverter
- Built-in charging
- Safe

Disadvantages

- Expensive
- Larger footprint
- Less customizable

4. Evaluation Criteria

Criterion	Weight
Initial Cost	25%
Runtime	20%
Weight	15%
Reliability	15%
Expandability	10%
Ease of Integration	10%
Maintainability	5%

The evaluation criteria were weighted according to project priorities. Initial acquisition cost received the highest weighting (25%) because minimizing prototype cost was identified as the primary design driver. Runtime (20%) and reliability (15%) were prioritized to ensure field usability. Weight, expandability, ease of integration, and maintainability were assigned lower weights because they represent desirable characteristics that can be improved through future design iterations.

5. Decision Matrix

Criteria	Weight	OEM Battery	SLA	LiFePO ₄	Power Station
Cost	0.25	3	5	2	2
Runtime	0.20	2	4	5	5
Weight	0.15	5	2	5	3
Reliability	0.15	4	5	5	5

Expandability	0.10	2	4	5	4
Integration	0.10	5	4	4	5
Maintainability	0.05	3	5	5	5

Alternatives were evaluated using a five-point ordinal scale, where 1 represents poor compliance with the criterion and 5 represents excellent compliance.

6. Weighted Scores

Alternative	Total Score
OEM Battery	3.15
SLA	4.35
LiFePO ₄	4.20
Power Station	3.75

7. Discussion

The Sealed Lead Acid (SLA) battery achieved the highest weighted score among the evaluated alternatives. Although heavier than the LiFePO₄ alternative, resulting in reduced portability during dismounted field operations, the SLA battery provides the lowest acquisition cost while satisfying the required two-hour operational endurance. Furthermore, the selected architecture supports future integration of a solar charging subsystem and can be upgraded to a LiFePO₄ battery with minimal modification to the electrical power distribution system. To mitigate the increased system weight, future design enhancements such as the incorporation of all-terrain transport wheels into the equipment case may be implemented to improve mobility over uneven terrain while maintaining overall system portability.

The LiFePO₄ alternative offers superior energy density, reduced weight, longer service life, and improved cycle life. However, it was not selected for the initial prototype because its significantly higher acquisition cost exceeds the project's current budget constraints. As a result, the SLA battery represents the most cost-effective solution while maintaining compliance with the operational requirements established for the TOMTOM Ground Control Station. The modular nature of the selected power architecture preserves a straightforward upgrade path should future funding permit adoption of a LiFePO₄ battery system.

8. Decision

Selected Alternative: 12V Sealed Lead Acid Battery

Rationale

The SLA battery provides the best balance between:

- Cost
- Runtime
- Reliability
- Ease of implementation
- Upgrade potential

The selected architecture preserves compatibility with future LiFePO₄ battery upgrades while minimizing prototype development cost. Additionally, the architecture satisfies the portability, endurance, maintainability, and modularity objectives established for the TOMTOM Ground Control Station and provides a scalable foundation for future system expansion. Based on the weighted evaluation and supporting technical analysis, the 12 V Sealed Lead Acid battery is recommended as the baseline power architecture for the initial TOMTOM Ground Control Station prototype. Future design iterations should reevaluate LiFePO₄ battery technology as budget constraints are relaxed, operational endurance requirements increase, or additional subsystem power demands are introduced.